

Infrastructure Resilience Proposal

To: CEO **From:** IT Department **Date:** April 1, 2026 **Subject:** High Availability & Disaster Recovery Plan for 800 CarGuru System

Background

On April 1, 2026, our primary production server (SMF_NEW_CG) experienced an unexpected shutdown, causing system downtime. The root cause is under investigation via AWS CloudTrail. This incident highlighted a critical gap in our infrastructure — **we currently have no failover mechanism.**

The Problem

- Our entire system runs on a **single server**
- If that server goes down, **all operations stop** — CRM, invoicing, customer management, WhatsApp, everything
- Recovery is **manual** and takes **15-30 minutes** at best
- There is **no real-time backup** of the database — we rely on periodic manual dumps

Proposed Solution

We propose implementing an **Active-Passive Failover System** with the following components:

1. Backup Server (Standby)

A secondary server that mirrors the main server, staying idle until needed. If the main server fails, it **automatically activates within 1-2 minutes** with no manual intervention.

2. Real-Time Database Replication

Every database change on the main server is **instantly replicated** to the backup server. No data loss in the event of a failure.

3. Automatic Failover

A monitoring script checks the main server every 60 seconds. If it detects failure, it automatically:

- Switches traffic to the backup server
- Activates the backup database
- All within **1-2 minutes** — no human action needed

4. Daily Backups

Automated daily database backups stored both on the server and downloaded to local storage for additional safety.

Business Impact

Scenario	Current	With Failover
Server goes down	Full outage, 15-30 min recovery	Auto-switch, 1-2 min downtime
Database corruption	Restore from last manual backup (hours of data loss)	Switch to replica (zero data loss)
Accidental deletion	No protection	Replica + daily backups
Staff needed for recovery	IT must be available immediately	Fully automated

Investment

Option	Monthly Cost	Notes
Cost-Saving Setup (recommended)	~\$100/month additional	Smaller backup instance (t3.large) — sufficient for temporary traffic
Full Mirror Setup	~\$280/month additional	Identical backup instance (t3.2xlarge)

Current infrastructure cost: ~\$280/month Proposed total: ~\$380-560/month depending on option chosen

Risk of Inaction

- **Revenue loss** during every outage — staff unable to process customers, create invoices, or manage leads
- **Customer dissatisfaction** — service delays and communication gaps
- **Data loss risk** — without real-time replication, any server failure could result in hours of lost transactions
- **Reputation damage** — repeated outages erode trust

Recommendation

We recommend proceeding with the **Cost-Saving Setup** immediately. It provides full automatic failover protection at a modest additional cost of ~\$100/month — a small price compared to the cost of even one hour of downtime.

A detailed technical implementation guide has been prepared and is ready for execution upon approval.

Next Steps

1. **Approval** to proceed with backup server setup
2. **Timeline:** Can be fully operational within **2-3 days**
3. **Testing:** Monthly failover drills to ensure reliability